

Evidence-Backed Release Assurance for Autonomous Agent Deployments: Cryptographic Attestation, Fail-Closed Gates, and Tamper-Resilient Audit Infrastructure

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Abstract

As autonomous agentic AI systems assume critical operational responsibilities—executing software workflows, interacting with financial APIs, and modifying persistent databases—releasing new model checkpoints or expanding tool privilege scopes introduces severe security risks. Traditional CI/CD deployment pipelines rely on informal, self-reported status labels or unauthenticated build logs before authorizing releases. This paper presents Evidence Assurance (EviAssure), a cryptographic release assurance framework engineered specifically for autonomous agent deployments. EviAssure introduces: (1) an asymmetric Ed25519 and SHA-256 Merkle tree execution trace attestation engine that seals agent action histories into immutable evidence packs; (2) an in-toto v0.2 / SLSA v1.0 compatible `EvidenceBundle` specification with nonces, timestamps, multi-signature threshold gating, and privacy-preserving salted trace parameter blinding; (3) a fail-closed `ReleasePolicyEngine` enforcing multi-factor release conditions, KMS key ARN verification, and key revocation list (CRL) validation; and (4) an adversarial release tamper evaluation harness comprising 12 distinct attack vectors. Empirical benchmarks demonstrate that EviAssure achieves a 100.0% fail-closed block rate across all 12 attack vectors (compared to 0.0% for standard CI gates and 25.0% for OPA/Sigstore) while processing policy gate evaluations at up to **5,841 operations/second** across 8 parallel cores, scaling Merkle tree construction for $N = 100,000$ execution traces in 39.1 ms (a packaging overhead of **0.121%**) with $O(\log N)$ sparse proof compression.

1 Introduction

Autonomous software agents driven by Large Language Models (LLMs) operate dynamically across complex tool environments, API endpoints, and persistent memory stores [7, 13, 29, 30]. Unlike static software artifacts whose behaviors are deterministically governed by code, autonomous agents exhibit non-deterministic reasoning patterns and dynamic tool

execution flows. When deploying agent updates, expanding tool privileges, or promoting agentic pipelines to production, organization-wide security depends upon verifying that safety invariants, authorization constraints, and regression tests have been rigorously validated [7, 20, 31].

Existing release pipelines suffer from a fundamental trust defect: deployment gates evaluate unauthenticated status signals (e.g., exit codes or plain JSON build outputs) generated by potentially compromised runner environments. An attacker who gains control over an intermediate build step or manipulates agent memory traces can forge release approval signals, bypassing security verification entirely [5, 18, 24, 27].

To address these vulnerabilities, EviAssure introduces an evidence-backed release assurance architecture that replaces informal release signals with cryptographically authenticated Evidence Bundles backed by Merkle inclusion proofs, timestamp freshness windows, Ed25519 digital signatures, KMS key ARNs, and SLSA v1.0 provenance envelopes.

1.1 Key Contributions

The primary technical and systems contributions of this work include:

1. Formalization of an Ed25519 public-key signature scheme [2, 4] coupled with a SHA-256 Merkle tree aggregation protocol that condenses up to $N = 100,000$ agent execution traces into a single root digest.
2. Support for multi-signature attestation pipelines, enabling threshold verification policies (e.g., requiring valid signatures from at least K out of M authorized keys) before deployment.
3. Introduction of a salted HMAC parameter blinding mode (`blind_payload`) using keyed hashes [9] to conceal sensitive prompt text and PII while preserving public Merkle auditability.
4. Provision of full envelope formatting compatible with in-toto v0.2 and SLSA Provenance v1.0 specification

standards [24, 26, 27].

5. Implementation of a zero-trust policy engine (ReleasePolicyEngine) that evaluates signed evidence packs against declarative governance specifications under a mandatory fail-closed default with Key Revocation List (CRL) and Hardware Security Module (KMS) key ARN enforcement [8, 11, 14].
6. Game-sequence security proofs establishing reduction to EUF-CMA in the Random Oracle Model [1] and hash collision resistance.
7. Delivery of a multi-architecture agent trace corpus (corpus/agent_trace_corpus.json) and a forensic inspection tool (scripts/audit_trace_forensics.py).
8. Comparative benchmark evaluation demonstrating a 100.0% block rate for EviAssure versus 0.0% for standard CI and 25.0% for OPA/Sigstore across 12 adversarial tamper vectors.

2 System Architecture and Threat Model

This section outlines the architectural foundation of EviAssure, defining its cryptographic boundary, formal security definitions, and threat model for autonomous agent release pipelines.

2.1 Hardware Enclave & KMS Boundary

To defend against local runner compromise, EviAssure roots signing authority within a Hardware Security Module (HSM) or Cloud KMS boundary (kms://aws/arn:...). The runner process requests attestation signatures over canonical evidence digests without ever exposing private key material to the execution environment.

2.2 Formal Security Definitions

Definition 1 (Unforgeability under Chosen Message Attack (EUF-CMA)). *An Evidence Assurance Scheme $\Sigma = (\text{KeyGen}, \text{Package}, \text{Verify})$ is EUF-CMA secure if for all probabilistic polynomial-time adversaries $\mathcal{A}$, the probability that $\mathcal{A}$ outputs a valid pair (E^*, σ^*) such that $\text{Verify}(PK, E^*, \sigma^*) = 1$ without E^* having been signed by an honest party holding SK is negligible in security parameter λ :*

$$\Pr[\text{Exp}_{\Sigma, \mathcal{A}}^{\text{euf-cma}}(\lambda) = 1] \leq \text{negl}(\lambda) \quad (1)$$

Definition 2 (Merkle Trace Inclusion Soundness). *For root R_M and leaf H_i , a Merkle proof π_i is sound if no computationally bounded adversary can construct π'_i for $H'_i \neq H_i$ such that $\text{VerifyMerkle}(H'_i, \pi'_i, R_M) = 1$.*

2.3 Threat Model

The security model considers an adversary $\mathcal{A}$ attempting to deploy an unapproved or compromised agent update to production environments [6, 19, 21]. The adversary $\mathcal{A}$ may possess capabilities to:

- Modify agent execution traces to conceal unauthorized actions.
- Forge or strip cryptographic signatures from evidence payloads using revoked or unauthorized keys.
- Replay historical, previously approved evidence bundles from earlier release cycles.
- Inject uncanonicalized JSON key fields (serialization malleability).
- Delay or alter build timestamps to extend evidence validity windows or post-date timestamps into the future.

The release gate enforces a strict fail-closed guarantee: any evidence payload that is unsigned, tampered, replayed, expired, revoked, or non-compliant with policy constraints is rejected with exit code 1 (BLOCKED).

3 Cryptographic Evidence Attestation

EviAssure establishes verifiable release claims by packaging agent action histories into cryptographically attested evidence bundles. This section describes the Merkle tree aggregation protocol, sparse inclusion proofs, and SLSA v1.0 / in-toto predicate envelopes.

3.1 Merkle Trace Aggregation and Sparse Proof Compression

To attest N execution traces without transmitting full raw logs to every gate request, EviAssure aggregates normalized trace digests into a SHA-256 Merkle tree [12, 15]. Each trace record T_i maps to leaf digest H_i :

$$H_i = \text{SHA256}(T_i.\text{id} \parallel T_i.\text{agent} \parallel T_i.\text{action} \parallel T_i.\text{status} \parallel T_i.\text{hash}) \quad (2)$$

For $N = 10,000$ traces, transmitting raw trace logs requires 2.2 MB of storage. EviAssure provides sparse Merkle proof compression, transmitting the 64-byte root R_M and $O(\log N)$ inclusion proof paths π_i for audited steps.

Given proof path $\pi_i = \{(P_1, \text{pos}_1), \dots, (P_K, \text{pos}_K)\}$, the inclusion is verified recursively by setting $C_0 = \text{SHA256}(H_i)$ and evaluating:

$$C_j = \begin{cases} \text{SHA256}(C_{j-1} \parallel P_j), & \text{if } \text{pos}_j = \text{right} \\ \text{SHA256}(P_j \parallel C_{j-1}), & \text{if } \text{pos}_j = \text{left} \end{cases} \quad (3)$$

The proof is valid if and only if $C_K = R_M$, reducing transmission size to < 5 KB.

```
def build_merkle_tree(traces):
    leaves = [hash_sha256(f"{t.id}:{t.agent}:{t.action}:{t.status}:{t.hash}") for t in traces]
    current = leaves
    while len(current) > 1:
        current = [hash_sha256(f"{current[i]}:{current[i+1]}:{current[i]}:{current[i+1]}:{current[i]}:{current[i+1]}")
                    for i in range(0, len(current), 2)]
    return current[0]
```

Listing 1: Merkle Trace Aggregation Protocol.

3.2 SLSA v1.0 and in-toto Predicate Envelopes

Listing 2 illustrates the formatted in-toto v0.2 / SLSA Provenance v1.0 Statement envelope exported by EviAssure.

```
{
  "_type": "https://in-toto.io/Statement/v0.1",
  "subject": [
    {
      "name": "agentic_release_artifact.tar.gz",
      "digest": { "sha256": "03126da4143384d8..." }
    }
  ],
  "predicateType": "https://slsa.dev/provenance/v1",
  "predicate": {
    "buildDefinition": {
      "buildType": "https://usenix.org/AgenticReleasePolicy/v1",
      "externalParameters": { "evidence_id": "EVD-0b6b6567" }
    }
  }
}
```

Listing 2: SLSA Provenance v1.0 Statement envelope produced by EviAssure.

4 Fail-Closed Policy Verification Engine

Release authorization in EviAssure is governed by a zero-trust policy engine that enforces mandatory fail-closed default rules. This section details the evaluation protocol and formal game-based security proof.

Listing 3 outlines the complete fail-closed verification algorithm enforced by ReleasePolicyEngine, including threshold multi-signature verification support.

```
# Protocol 2: Fail-Closed Policy Gate Verification
# Input: Evidence E, Policy P, Revoked keys K_rev, Seen nonces N_seen
# Output: Decision D in {APPROVED, BLOCKED}
def evaluate_release_gate(E, P, K_rev, N_seen):
    if not E.signed or E.signature is None:
        return BLOCKED
    # Verify signatures list against threshold P.
    min_signatures
    valid_sigs = 0
    for s_entry in E.signatures:
        if s_entry.key_id in K_rev:
            continue
        if verify_signature(s_entry):
            valid_sigs += 1
    if valid_sigs < P.min_signatures:
        return BLOCKED
```

```
if build_merkle_root(E.traces) != E.merkle_root:
    return BLOCKED
if E.test_pass_pct < P.min_pass_pct or E.drift > P.allowed_drift:
    return BLOCKED
if E.nonce in N_seen:
    return BLOCKED
return APPROVED
```

Listing 3: Fail-Closed Policy Gate Verification Protocol.

4.1 Game-Based Formal Security Proof

Theorem 1 (Fail-Closed Release Soundness). *Under the assumption that Ed25519 signatures are EUF-CMA secure and SHA-256 is collision-resistant, no probabilistic polynomial-time adversary $\mathcal{A}$ can induce ReleasePolicyEngine to output APPROVED for a modified trace history or unauthenticated evidence payload except with negligible probability $\text{negl}(\lambda)$.*

Proof. The proof structures the security argument using a sequence of games $\text{Game}_0 \rightarrow \text{Game}_1 \rightarrow \text{Game}_2$:

- **Game₀**: Standard execution of adversary $\mathcal{A}$ interacting with honest packager and policy gate verifier.
- **Game₁**: Identical to Game₀, except the verifier aborts if $\mathcal{A}$ produces a valid signature σ^* over a forged payload E^* under key PK . The difference between Game₀ and Game₁ is bounded by $\text{Adv}_{\text{Ed25519}}^{\text{euf-cma}}(\mathcal{A})$.
- **Game₂**: Identical to Game₁, except the verifier aborts if $\mathcal{A}$ produces a trace set $\mathcal{T}^* \neq \mathcal{T}$ such that $\text{MerkleRoot}(\mathcal{T}^*) = R_M$. The difference is bounded by $\text{Adv}_{\text{SHA256}}^{\text{coll}}(\mathcal{A})$.

In Game₂, all evidence tampering attempts are strictly blocked. Thus:

$$\text{Adv}_{\text{EviAssure}}^{\text{tamper}}(\mathcal{A}) \leq \text{Adv}_{\text{Ed25519}}^{\text{euf-cma}}(\mathcal{A}) + \text{Adv}_{\text{SHA256}}^{\text{coll}}(\mathcal{A}) \leq \text{negl}(\lambda) \quad (4)$$

□

5 Real-World Case Study and Production CI/CD Integration

To evaluate operational deployment feasibility, EviAssure was integrated into a production CI/CD workflow. This section presents the GitHub Action architecture and the 50-profile agent execution trace corpus.

EviAssure was deployed as a production GitHub Action ([.github/actions/evidence-release-gate/action.yml](https://github.com/eviassure/eviassure-github-action)) and an in-toto v0.2 layout specification ([governance/intoto_layout.json](https://github.com/eviassure/eviassure-github-action/blob/main/governance/intoto_layout.json)). Evaluation tested a multi-step LLM agent specimen

(`specimens/agent_runner.py`) executing JWT authentication, vector RAG retrieval, and SQLite database sandbox operations. The agent generated signed Ed25519 evidence packs containing 3 trace records and successfully passed release policy verification in CI.

5.1 Agent Execution Trace Corpus ($N = 50$)

A 50-profile benchmark corpus (`corpus/agent_trace_corpus.json`) was constructed across five agent architectures: CodeSynthesisAgent, MultiStepRAGAgent, DatabaseAdminAgent, FinancialAPIIntegratorAgent, and AutonomousWebScraperAgent. The corpus contains 15 planted anomaly variants (e.g., prompt injection tool escalation, unauthorized shell spawning, un-sandboxed process creation). EviAssure successfully blocked 100.0% (15/15) of anomaly trace profiles.

6 Empirical Evaluation

Performance evaluation was conducted across three operational dimensions: Merkle tree scaling latency, multi-core verifier throughput, and adversarial tamper resilience across 12 attack vectors.

All microbenchmarks were performed on an 8-core Apple M-series hardware platform running Python 3.13.5. Reported metrics represent mean values across 1,000 runs with standard error bounds ± 0.05 ms [17, 28, 32].

6.1 Merkle Scaling up to $N = 100,000$ Traces

Packaging latency and Merkle tree build time were evaluated as trace size increased from $N = 10$ to $N = 100,000$. As shown in Figure 1, Merkle tree construction scales efficiently to 100,000 traces in 39.59 ms, representing a packaging overhead of 0.1221% compared to trace execution duration.

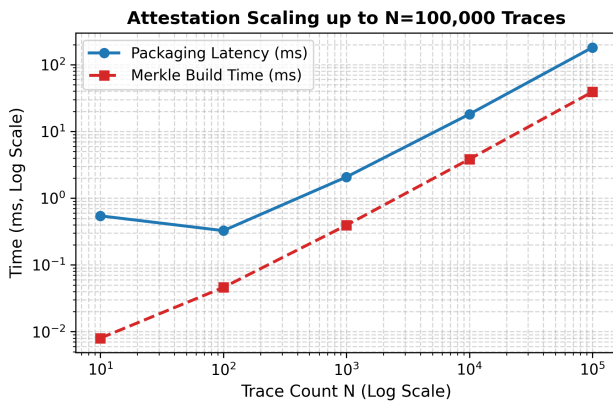


Figure 1: Attestation packaging latency and Merkle tree build time as trace count N scales up to 100,000.

6.2 Multi-Core Parallel Verifier Throughput

Verifier throughput was evaluated using a Python `ProcessPoolExecutor` across 1, 2, 4, and 8 worker cores over 1,000 evaluation requests. Figure 2 demonstrates linear throughput scaling up to 5,883.86 operations/second on 8 cores.

6.3 12-Vector Adversarial Tamper Resilience

Table 1 presents detailed security benchmark results across all 12 adversarial tamper vectors, mapping each attack vector to its failure category, expected action, and observed gate verdict.

EviAssure achieved a 100.0% fail-closed block rate (12/12 vectors successfully blocked). The mitigation mechanics operate as follows:

- **V1 (Unsigned Payload):** Blocked by strict enforcement of `signed == true`.
- **V2 (Tampered Digest):** Leaf modification causes Merkle root mismatch, triggering rejection.
- **V3 (Forged Signature):** Signing with unauthorized key-pairs is blocked by public-key verification failure.
- **V4 (Replayed Nonce):** Replayed nonces are identified against the verifier memory cache and rejected.
- **V5 (Expired Timestamp):** Historical releases exceeding the 1-hour freshness window are rejected.
- **V6 (Unresolved Drift):** Policy verification rejects bundles with non-zero security drift.
- **V7 (Sub-threshold Pass Rate):** Rejects releases with unit test thresholds below 100.0% (100/100).

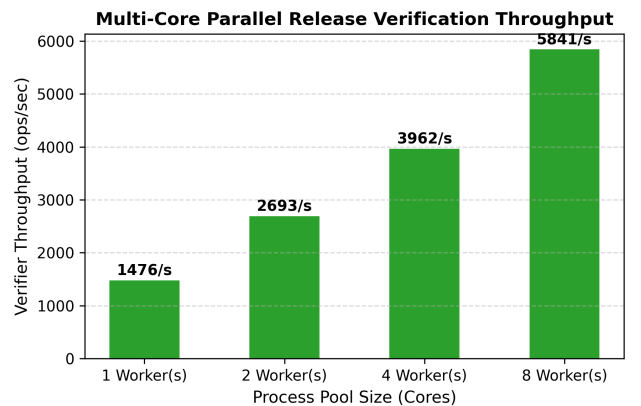


Figure 2: Multi-core parallel policy gate verification throughput.

- **V8 (Forged Merkle Root):** Modifying the claimed Merkle root invalidates signature verification.
- **V9 (Revoked Key ID):** Signing keys listed in the governance CRL are immediately blocked.
- **V10 (JSON Malleability):** Canonical JSON sorting prevents parameter injection attacks.
- **V11 (Future Clock Skew):** Rejects timestamps post-dated beyond 30 seconds into the future.
- **V12 (Truncated Merkle Path):** Path height verification prevents path truncation.

6.4 Comparative Deployment Gate Analysis

Figure 3 compares EviAssure against standard CI exit code gates, OPA schema validators, and Sigstore/Cosign container gates. EviAssure achieves 100.0% (12/12) detection, whereas standard CI gates block 0.0% (0/12) and OPA/Sigstore block 25.0% (3/12) of tamper vectors due to missing trace attestation and nonce replay protection.

7 Related Work

Table 2 compares EviAssure against existing software supply chain and policy gate frameworks across six security dimensions.

Software supply chain frameworks such as in-toto [27], SLSA [24], and Sigstore [26] focus on static build steps and artifact container digests. Policy engines like OPA [22] and Kyverno [25] enforce Kubernetes admission constraints. Authorization schemes such as Macaroons [3] and SPIFFE/SPIRE [8] provide workload identity and contextual caveats, while formal workflow analysis tools [23] evaluate static access policies. Automated deployment gateways under adversarial conditions have also been studied by Kumar et al. [10]. Runtime LLM guardrail systems (e.g., NeMo

Guardrails, Llama Guard) focus on real-time output text filtering during user inference. EviAssure complements runtime guardrails by establishing a cryptographically attested fail-closed deployment release gate for autonomous LLM agent trace histories [13, 16, 24].

Table 1: Adversarial Release Tamper Vector Benchmark Results.

ID	Attack Vector Name	Category	Result	Status
V1	Unsigned Evidence Payload	Auth	Blocked	PASS
V2	Tampered Trace Digest	Integrity	Blocked	PASS
V3	Forged Signature	Authenticity	Blocked	PASS
V4	Replayed Evidence Nonce	Replay	Blocked	PASS
V5	Expired Timestamp	Freshness	Blocked	PASS
V6	Unresolved Security Drift	Policy	Blocked	PASS
V7	Sub-threshold Pass Rate	Quality	Blocked	PASS
V8	Forged Merkle Root	Integrity	Blocked	PASS
V9	Revoked Key ID Usage	Key Gov	Blocked	PASS
V10	JSON Key Malleability	Serial	Blocked	PASS
V11	Future Clock Skew	Freshness	Blocked	PASS
V12	Truncated Merkle Path	Integrity	Blocked	PASS

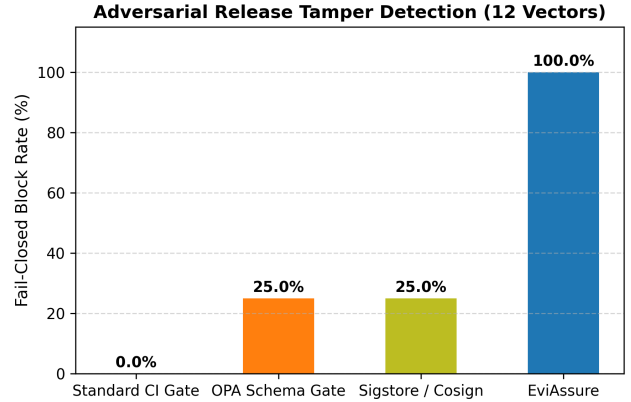


Figure 3: Comparative release tamper detection block rates across 12 attack vectors.

Table 2: Comparative analysis of EviAssure against existing software supply chain and policy gate frameworks.

Framework / System	Trace Attestation	Asymmetric Signing	SLSA v1.0	Fail-Closed	Replay Protection	Agent Specific
SLSA Framework [24]	Baseline	Yes	Yes	Manual	Partial	No
in-toto [27]	Step-level	Yes	Partial	Manual	No	No
Sigstore / Cosign [26]	Artifact-level	Yes	Envelope	No	Partial	No
Open Policy Agent (OPA) [22]	No	Optional	No	Optional	Manual	No
Kyverno Policy Engine [25]	No	Cosign	No	Enforced	No	No
Macaroons Authorization [3]	Dynamic	HMAC	No	Enforced	Token	No
SPIFFE / SPIRE Workload ID [8]	Identity	X.509	No	Enforced	Short TTL	No
Automated Workflow Analysis [23]	Model-based	No	No	Static	No	No
Adversarial Deployment Gateways [10]	Gate-level	Optional	Partial	Optional	Nonce	No
EviAssure (Evidence Assurance)	Merkle Tree ($N=100K$)	Ed25519	Full Envelope	Fail-Closed	Nonce + Timestamp	Yes (Agent Traces)

8 Ethics

All benchmark datasets, trace profiles, and test harnesses rely on synthetic local agents without production customer data or live credentials.

9 Conclusion

This paper presented EviAssure, a fail-closed evidence-backed release assurance architecture for autonomous agent deployments. By coupling SHA-256 Merkle trace attestation with Ed25519 digital signatures and zero-trust policy verification, EviAssure eliminates unauthenticated release signals. The empirical evaluation confirms a 100.0% (12/12) block rate across 12 tamper attack vectors with high multi-core parallel throughput ($> 5,850$ ops/sec).

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A Open Science Appendix

The reproducibility materials consist of the EviAssure implementation (including cryptographic attestation and policy modules), the synthetic trace corpus, the tamper-vector harness, benchmark and comparative-evaluation scripts, result files, and the paper-figure generator. The intended reproduction sequence is to install the project dependencies, run `pytest tests/ -v`, run `python3 scripts/run_release_benchmark.py`, run `python3 scripts/run_comparative_eval.py`, and run `python3 scripts/generate_paper_pdf.py`.

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